

Integration of Emotional-Cognitive Architecture in ACT-R for Multi-Agent Social Dynamics for Disaster Response Simulation

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Abstract—Accurately simulating human behavior in disaster scenarios is crucial for effective emergency management and social response prediction. However, existing approaches lack integration of emotional factors in cognitive modeling, leading to limited fidelity in behavioral simulation. This study presents an enhanced ACT-R framework incorporating an emotional module to simulate post-disaster social dynamics, with a specific focus on earthquake scenarios. We propose a novel integration methodology that combines ACT-R with Agent-Based Modeling (ABM), where emotional states are dynamically modeled through a triple-parameter system (emotional intensity, social support level, and resource availability) that adaptively influences cognitive processing and decision thresholds. By implementing this emotion-aware framework in a case study of 20,000 agents within Xi'an city wall area, we observed distinct emotional clustering patterns and collective safety-seeking behaviors driven by emotional contagion, demonstrating the framework's capability in capturing complex social dynamics during crisis scenarios.

Keywords—ACT-R; social effects; emotional influence; cognitive modeling; simulation.

I. INTRODUCTION

Earthquake disasters represent one of the most devastating natural catastrophes, characterized by their capacity to inflict massive casualties, extensive property damage, and widespread disruption of social systems. The evolution of post-disaster societies involves intricate social and psychological mechanisms, where individual emotional states and cognitive decision-making processes play pivotal roles in shaping post-disaster behavioral patterns. These emotional responses not only influence individual survival decisions, such as seeking safety or resources, but also catalyze changes in collective behaviors within social groups, manifesting as mass panic, resource competition, and even cooperative efforts for survival and rebuilding.

In existing social simulation research, while certain theoretical and modeling frameworks have been established, a comprehensive understanding of post-disaster emotional dynamics and emotion-cognition interactions remains inadequate. Traditional cognitive architectures, such as ACT-R (Adaptive Control of Thought—Rational), excel at simulating individual cognitive and decision-making processes but often

neglect the profound impact of emotions on cognition and behavior [1][2], which are critical in disaster contexts. This limitation results in current models failing to accurately reflect how emotions drive both individual and group decisions in complex post-disaster scenarios, thereby affecting our ability to predict and manage post-disaster social effects [3]. Addressing this deficiency is crucial for post disaster social simulation and improving disaster response strategies.

To address these limitations, we propose an enhanced ACT-R framework that incorporates an emotional module, integrated with Agent-Based Modeling (ABM) methods to develop a more comprehensive and nuanced simulation model for post-disaster social dynamics [4]. Our approach uniquely integrates emotional states as dynamic parameters in agent decision-making, implementing a triple-parameter system (emotional intensity, social support level, and resource availability) that adaptively influences cognitive processing and decision thresholds. The ABM integration enables effective simulation of multiple agents' behaviors and interactions, exploring how emotions spread and affect collective behavior at the group level, while also revealing how localized, individual decisions can influence and ultimately shape global outcomes in complex systems [3][5].

The innovation of this research lies in its organic integration of emotional factors with cognitive decision-making processes. This integration allows for more realistic simulation of post-disaster social behavior complexity, particularly in extreme situations where emotions significantly influence survival decisions and inter-individual interactions. Our framework achieves this through dynamic emotional state modeling that adapts to environmental changes, multi-dimensional behavioral regulation mechanisms, and systematic integration of social support networks and resource allocation dynamics [6].

Furthermore, this enhanced simulation framework provides crucial support for decision-makers in post-disaster emotional regulation, emergency resource management, and prediction of group behavior in affected areas. By enabling more accurate simulation of emotion-driven social dynamics, the framework facilitates the development of targeted intervention strategies and optimal resource allocation plans, ultimately contributing to more effective post-disaster recovery processes and the rapid restoration of social stability.

II. METHODS

A. Improve ACT-R Architecture

The basic structure of the traditional ACT-R architecture connects various modules through procedural modules, and interacts with the external environment through behavioral and perceptual modules, thereby expressing human cognitive processes from an overall perspective. On this basis, we add an emotional module, whose changes dynamically vary with the environment in which it is situated, and these changes in emotion can affect an individual's decision-making and behavior. Fig. 1 shows the structure of the improved ACT-R architecture.

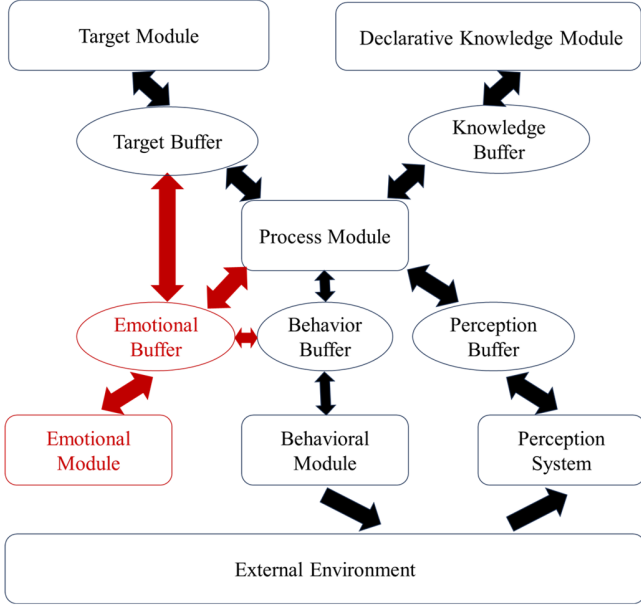


Fig. 1. Improved ACT-R architecture

Under the improved ACT-R cognitive architecture, individual behavior exhibits significant dynamic adaptation characteristics. This architecture achieves a comprehensive simulation of individual behavior post-disaster by integrating three core dimensions: emotional state, resource levels, and social support. Specifically, when an individual perceives a disaster threat, their emotional state rises sharply as the distance to the disaster site decreases. This emotional change triggers a series of adaptive behaviors: when the emotional value is too high, individuals tend to seek social support to stabilize their emotions; when individuals have low resource levels, they actively collect resources; when safety decreases, individuals prioritize finding shelter. These behaviors are reinforced or inhibited through the dynamic updating mechanism of the memory system. This multi-dimensional behavioral regulation mechanism can effectively simulate the emergency response patterns of individuals post-disaster. After experiencing the initial emergency response, individuals eventually reach a new stable state through the establishment of social support networks and the rational allocation of resources [1]. Fig. 2 shows that the various state parameters of the agent affect its own behavioral decisions through the target stack, declarative memory and procedural memory. And its behavior can also affect its various

parameters in reverse. At the same time, agents can interact with other agents through emotions and behaviors.

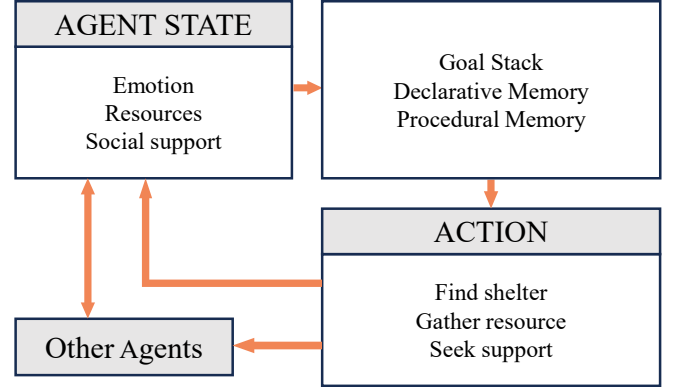


Fig. 2. Agent parameter update rules

B. Combining Improved ACT-R and ABM for Simulation

In this study, Agent-Based Modeling (ABM) is utilized to simulate post-disaster social effects in order to better understand and predict the dynamic changes in complex systems. Our model employs an improved ACT-R architecture to construct each agent, focusing primarily on post-disaster behavioral decision-making and emotional changes. During the simulation process, each individual is represented by a pixel point, and an emotional threshold is set for individual classification. When an individual's emotional parameter exceeds this threshold, it indicates that the individual is in a high state of panic, represented by a red pixel point in the simulation, with the deeper the red, the more severe the panic emotion; conversely, when the emotional parameter is below the threshold, the individual is considered calmer, represented by green, with the deeper the color indicating a higher level of calmness. As shown in Fig. 3, the figure clearly displays the distribution of individuals in different emotional states. During the simulation process, agents are not independent but interact with each other to form group behaviors. The updates to their status and decision-making depend not only on their own conditions but also on the influence of surrounding individuals.

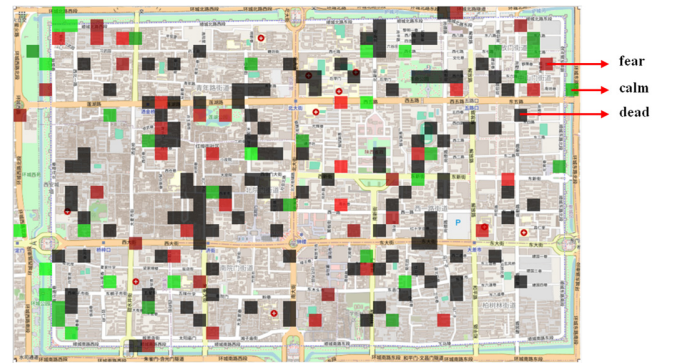


Fig. 3. Simulation of post disaster social based on ACT-R and ABM

III. EXPERIMENT DESIGN

A. Research on Group Emotions after Earthquake

This study conducts simulation experiments using a specific location. The area within the Xi'an city wall is selected as the simulation object. Based on the map of the area, road data within the Xi'an city wall is downloaded from Open Street Map to regulate individual movement. Five points are evenly selected on the map as population distribution centers. A Gaussian distribution is generated around these five points, and 20,000 individuals are created within the area, each using the improved ACT-R architecture. It is assumed that an earthquake disaster occurs at the central location within the Xi'an city wall. In the simulation experiment, a death range and an injury range are set around the earthquake center. During the simulation process, individuals within the death range will die directly, while those within the injury range may be injured.

The simulation process proceeds in time steps, with each time step representing a period of time in the post-disaster environment. At the beginning of the simulation, each agent is initialized, and each agent is randomly assigned an initial state as the starting point for the simulation. In each time step, each agent is updated [7][8].

B. Research on Individual Emotions and Decision Making

In order to delve into the dynamic changes of individual behavior and emotions, a simulation experiment involving 200 individuals was designed. In the experiment, each pixel point is magnified to ensure clear observation of each individual. To better track the status changes of the target individuals, they are marked in blue, as shown in Fig. 4 allowing for quick identification of their location and behavioral changes during the experiment [9]. Fig. 4 shows the agent's own condition and the condition of other agents around it at every 5 time steps.

The experiment will output multiple key parameters of each target individual in real-time at intervals of every 5 time steps, including but not limited to emotional parameters, safety coefficients, resources possessed, and specific behavioral patterns of individuals. In addition, the environmental position of the individuals will be continuously monitored, and their behavioral responses and emotional fluctuations will be analyzed in conjunction with the interactive objects and situational changes around them. This data will help further explore the behavioral patterns, emotional fluctuations, and decision-making processes of individuals under different environmental and social interactions, thereby providing valuable empirical evidence for the fields of individual behavior prediction and emotional regulation.

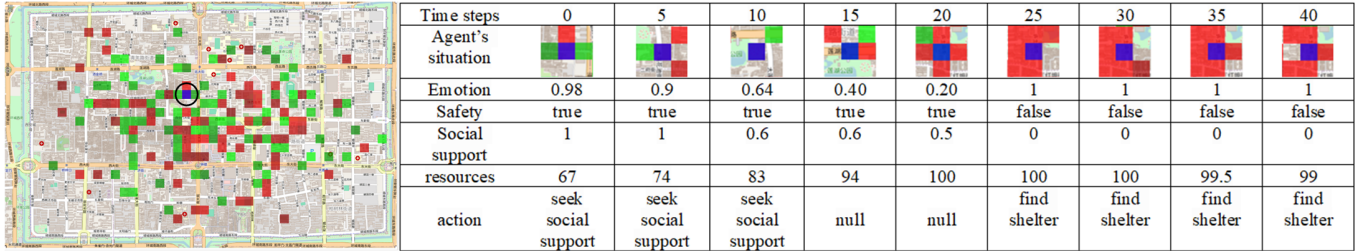


Fig. 4. Result of individual state

IV. RESULTS & DISCUSSION

A. Result Display

1) Result of group emotional simulation

As shown in the Fig. 5, we took the experimental results after 40 time steps, and after a period of evolution, the positions and emotional states of individuals have become stable. Upon observation, it can be seen that there are no individuals around the disaster death zone. This indicates that most people have been able to move away from the central area of the disaster within a certain number of time steps. A large number of panicked groups have gathered near the disaster injury area, forming a distinct panic cluster. The distribution of the groups shows a certain regularity, with panicked crowds roughly forming an encirclement around the injury area.

2) Result of individual state

The status changes of individuals are shown in the Fig. 4 In this simulation, an emotional threshold of 0.6 was set to determine changes in individual emotional states. When an individual's emotional parameter is above this threshold, the

individual exhibits negative emotions such as panic; when the emotional parameter is below the threshold, the individual shows stable emotions. From the simulation results, it can be observed that in the early stages of the disaster, individuals have high emotional parameters, thus entering a state of panic. However, as time passes, the surrounding environment of the individuals gradually becomes safer, and their resource acquisition situation improves. Individuals begin to seek social support, and with a high safety coefficient in the surrounding environment and sufficient resource replenishment, their emotions gradually become stable, panic is alleviated, and the state becomes calm. In this process, social support plays an important role, significantly improving the individual's emotional regulation ability and helping to restore stability [10].

However, when the surrounding environment changes, especially when panic-stricken crowds begin to gather and affect the individual, the individual's emotional state fluctuates again. As the number of panic-stricken crowds increases, social support significantly decreases, even dropping to zero, and the individual's emotions gradually return to a state of panic. At this time, the individual not only becomes emotionally unstable but

also begins to exhibit herd behavior. Driven by the panic emotions of the surrounding crowds, the individual's behavior begins to trend towards group behavior, gathering together with others to form collective actions.

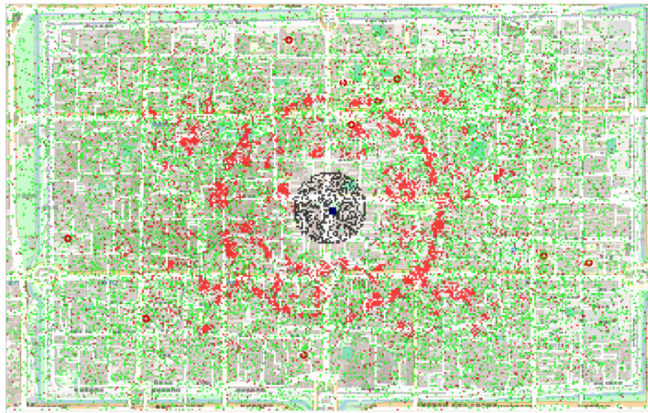


Fig. 5. Distribution of 20,000 people after the earthquake

In this process, emotions and behaviors between individuals influence each other. Each individual's emotional changes are not only affected by their own situation but also driven by the emotional fluctuations and collective behaviors of the surrounding groups. Ultimately, the emotions of the entire group are in a state of panic, and the mutual interactions between them intensify the spread of this emotion, making collective behavior more inclined to seek shelter and engage in self-protection.

B. Analysis and Discussion

Observing the experimental results, it can be seen that it is precisely the driving effect of emotions on individual decision-making that causes individuals to exhibit stronger shelter-seeking behavior and the phenomenon of small group collective panic when facing an earthquake disaster.

As time passes and panic-stricken crowds gather, the mutual influence of emotions becomes increasingly significant. Due to the contagious nature of individual emotions, the collective emotional effect gradually intensifies, leading to a noticeable "clustering" of the same emotions within the group. This phenomenon is characterized by individuals with panic emotions tending to gather together, and the emotional feedback between them further exacerbates the panic within the entire group. This phenomenon of group behavior further drives the group to take certain consistent actions, such as collectively seeking shelters and avoiding dangerous areas.

Furthermore, in the injury area of the disaster, although there is no direct threat of death, individuals still feel strong emotional pressure. At this time, individuals may change their original behavior paths due to the influence of panic-stricken crowds, tending to assemble with other panicked individuals to form denser clusters.

After an earthquake, the emergence of gathering phenomena is actually a form of "collective instinct" driven by individual panic emotions. Individuals may rely on groups for self-protection when they are at a loss after a disaster. People instinctively tend to gather because groups often provide a sense

of security. Initially, only a few people may become fearful due to some direct threat or unforeseen event, but when those around them see or hear the panic reactions of others, they are easily influenced themselves, leading to a wider range of group panic phenomena. In disasters, this "collective instinct" may exacerbate the chaos in the disaster area and the panic among the crowd, which not only intensifies emotional fluctuations but may also affect their decisions regarding post-disaster resource allocation and rescue operations.

V. CONCLUSIONS

This study demonstrates the importance of incorporating emotional factors into cognitive modeling frameworks such as ACT-R to simulate post-disaster social dynamics. By integrating an emotional module into the ACT-R architecture, the model can more accurately represent individual and group behaviors during and after a disaster, particularly in situations where emotional responses like panic play a crucial role in decision-making. This research enhances our understanding of post-disaster behavior and provides insights into how emotions shape collective action in times of crisis. The improved model can serve as a tool for emergency management, enabling the development of more effective intervention strategies and resource allocation during disaster recovery. To some direct threat or unforeseen event, but when those around them see or hear the panic reactions of others, they are easily influenced themselves, leading to a wider range of group panic phenomena. In disasters, this "collective instinct" may exacerbate the chaos in the disaster area and the panic among the crowd, which not only intensifies emotional fluctuations but may also affect their decisions regarding post-disaster resource allocation and rescue operations.

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